

Practical 2: Understanding a System in Motion

State, Situation, Events, Behaviour, Process, and Function

1. Why this Practical?

In Practical 1, you learned to recognize a system and identify its purpose, stakeholders, components, interactions, boundary, environment, and emergent behaviour.

However, a system cannot be understood only from its components or from a single observation. Its condition changes with time, events, operating context, stakeholder actions, and environmental influences. The same apparent system state may also have different operational significance in different situations. In this practical, you will examine one system dynamically—how it changes, behaves, performs its functions, and responds to events.

2. Learning Objectives

After completing this practical, you should be able to:

- distinguish among the state, situation, behaviour, process, and function of a system;
- identify variables that describe the state of a system;
- recognize how context and relationships transform a state into a situation;
- trace changes in the system following significant events;
- distinguish what a system does from how it does it;
- identify intended, unintended, and emergent behaviour; and
- analyze how a change in one part of a system affects its overall functioning.

3. Group and System Selection

Continue with the same group of five students formed for Practical 1.

Select **one sufficiently complex system** from the five systems studied in Practical 1. Prefer a system:

- involving people, technology, rules, and an operating environment;
- whose condition changes with time;

- in which decisions or interventions are made; and
- for which the group can observe or reasonably reconstruct a sequence of events.

Do not select a simple object whose behaviour can be described entirely through its physical components.

4. Practical Tasks

A. Re-establish the System

Briefly present:

1. system name and purpose;
2. principal stakeholders;
3. system boundary;
4. important environmental elements; and
5. the reason for selecting this system.

This section should be based on Practical 1 but refined wherever necessary. Do not reproduce the earlier analysis in full.

B. Identify the State of the System

Identify **six to eight variables** that collectively describe the condition of the system at a particular instant.

Prepare the following table:

State variable	Possible values or conditions	How it can be observed or estimated	Why it matters
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Include different types of variables, wherever applicable:

- physical or technical;
- human;
- informational;
- organizational; and
- environmental.

Then describe **three distinct states** of the system:

1. normal operating state;
2. stressed or degraded state; and
3. failed, recovering, or unusually successful state.

C. Construct Situations from States

A state describes the condition of the system at an instant. A situation includes that state together with the relevant context, relationships, history, stakeholder concerns, and likely consequences.

Construct **two situations having substantially the same observable state but different operational significance**.

For each situation, identify:

- the common observable state;
- the surrounding context;
- relevant stakeholder relationships;
- preceding events or history;
- stakeholder concerns;
- possible consequences; and
- why the required response may be different.

The differences must arise from context or relationships—not merely by changing the values of the state variables.

D. Trace Events and State Changes

Identify a meaningful operating episode involving **at least five events**. The events may be routine, disruptive, corrective, or externally initiated.

Represent the episode using a state-transition diagram or an annotated timeline.

For every transition, state:

Previous state	Triggering event	Resulting state	Stakeholder affected	Immediate consequence
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Also identify:

- one event that the system can anticipate;
- one event that is difficult to anticipate; and
- one event that appears minor but can produce a significant system-level effect.

E. Analyze System Behaviour

Using the episode studied in Part D, analyze how the system behaves over time.

Identify:

1. one intended behaviour;
2. one unintended behaviour;
3. one adaptive or corrective behaviour;

4. one emergent behaviour, if present; and
5. one behaviour that may be interpreted differently by different stakeholders.

Support every identified behaviour with observations or evidence from the chosen episode. Do not merely list desirable or undesirable characteristics.

F. Distinguish Function from Process

Identify **three important functions** performed by the system.

For each function, state:

- what outcome the function is intended to produce;
- which stakeholder requires it; and
- how its successful performance can be recognized.

Select one function and construct its corresponding process diagram showing:

- initiating condition;
- activities;
- participating components or stakeholders;
- information or material exchanged;
- decision points;
- expected output; and
- possible points of disruption.

Then answer:

1. Can the same function be achieved through a different process? Illustrate with one alternative.
2. Can the process be completed while the intended function remains unfulfilled? Construct one plausible example.

G. Introduce a Change in the System

Propose one realistic change in a component, rule, resource, stakeholder action, technology, or environmental condition.

Trace its effects on:

- system state;
- stakeholder relationships;
- process execution;
- system behaviour;
- intended function; and
- possible unintended consequences.

Conclude whether the change improves the **system as a whole** or only one part of it.

H. Group Synthesis

Prepare a one-page synthesis addressing:

- What could be understood only after observing the system over time?
- Which distinction—state and situation, behaviour and process, or process and function—was most difficult to apply?
- What important aspect would have remained unnoticed if the system had been examined only through its components?

I. Individual Reflection

Each student should independently answer, in **not more than 250 words**:

How has studying the system in motion changed my understanding of what it means to analyze a system?

The reflection must include one specific insight obtained from the group's system analysis.

5. Deliverables

Each group shall submit:

Code	Deliverable
A	Refined system description, purpose, boundary, and selection rationale
B	State-variable table and three system states
C	Two contextually different situations having substantially the same observable state
D	Event-based state-transition diagram or annotated timeline
E	Analysis of intended, unintended, adaptive, and emergent behaviour
F	Function analysis and detailed process diagram
G	System-wide analysis of one proposed change
H	One-page group synthesis
I	Individual reflection from every group member
J	Contribution statement indicating the work undertaken by each member

6. Submission Guidance

- Use diagrams and tables wherever they improve clarity.
- Base the analysis on observation, credible information, or a carefully constructed realistic episode.
- Clearly distinguish observed information from assumptions.
- Avoid treating the system as merely a collection of components.
- Do not use the terms **state, situation, behaviour, process, and function** interchangeably.
- The submission should demonstrate the group's reasoning, not merely reproduce definitions.
- A suggested length is **8–10 pages**, excluding individual reflections and references.

7. Suggested Reading

1. Ackoff, R. L. (1971). "Towards a System of Systems Concepts." *Management Science*, 17(11), 661–671.
2. Ackoff, R. L., and Emery, F. E. (1972). *On Purposeful Systems*. Aldine-Atherton.
3. Checkland, P. (1999). *Systems Thinking, Systems Practice*. Wiley.
4. Meadows, D. H. (2008). *Thinking in Systems: A Primer*. Chelsea Green Publishing.
5. INCOSE. *Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*.
6. Crawley, E., Cameron, B., and Selva, D. (2016). *System Architecture: Strategy and Product Development for Complex Systems*. Pearson.
